

A Project Report

On

“AI-Based Crypto-Currency Trading System”

By

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Bhaskar Waman Thakur College of Science,

Yashvant Keshav Patil College of Commerce,

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(VIVA College)

(Affiliated to University of Mumbai)

Virar West, 401303

Maharashtra

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**Vishnu Waman Thakur Charitable Trust's
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CERTIFICATE

Roll No. 36

This is to certify that the project entitled, “AI-Based Crypto-Currency Trading System”, is bonafied work of Rushabh Mahesh Sanghvi bearing submitted in partial fulfillment of the requirements for the award of degree of BACHELOR OF SCIENCE in DATA SCIENCE from University of Mumbai.

Internal Guide

Head of Department

External Examiner

Date:

College Seal

DECLARATION

I hereby declare that the project entitled, “**AI-Based Crypto-Currency Trading System**” done at “**Vishnu Waman Thakur Charitable Trust’s Bhaskar Waman Thakur College of Science, Yashwant Keshav Patil College of Commerce, Vidya Dayanand Patil College of Arts**”, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (DATA SCIENCE)** to be submitted as final semester project as part of our curriculum.

ACKNOWLEDGEMENT

I am very grateful to our principal for providing us with an environment to complete my project successfully.

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We would like to thank our parent for providing us with all their support and encouragement right from the project's budding stage to its current maturity.

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Abstract or Project Summary

1. Project Abstract — Bitcoin Price Prediction & Backtesting (Binance 1-min).

This project builds a **simple, end-to-end ML pipeline** to forecast short-term Bitcoin moves and evaluate trading performance. The focus is clarity over complexity: clean data ingestion, lightweight features, simple labeling, compact models, and an interpretable **Streamlit dashboard** for backtesting and (simulated) live trading.

2. Data.

We pull public **Binance BTCUSDT 1-minute candles** and persist them as `btc.csv` with columns: `time, open, high, low, close, volume`. A minimal extractor keeps the dataset reproducible and auditable.

3. Problem Framing.

Instead of predicting exact prices, we frame it as **3-class direction classification** over a fixed horizon: **SELL / HOLD / BUY**. A trade label is assigned by the **future return** over a horizon (H) (e.g., **60 minutes**) using simple thresholds (e.g., $\pm 0.5\%$):

- future return $> +\text{threshold}$ → **BUY**
- future return $< -\text{threshold}$ → **SELL**
- otherwise → **HOLD**

4. Features.

From OHLCV we derive a small set of classic technical features: short/medium **returns** (1, 5), **MA(5)–MA(20)** spread, **RSI(14)**, **MACD diff**, **Bollinger %B**, **ATR**, **OBV**, and a **volume z-score**. These are easy to compute, widely understood, and fast.

5. Models.

We compare **Logistic Regression**, **Random Forest**, **Gradient Boosting**, and **XGBoost** (final choice). XGBoost is compact, fast to train, handles non-linearities, and provides class probabilities for confidence gating. Class imbalance (few BUY/SELL vs many HOLD) is addressed via **balanced sample weights**.

6. Training & Evaluation.

We use chronological splits (no leakage). Metrics reported: **accuracy**, **precision/recall/F1** per class, and **confusion matrix** to reveal BUY/SELL vs HOLD behavior. Because trading results matter most, we add a **one-trade-at-a-time backtest** that enters on BUY/SELL,

holds exactly (H) minutes, exits, and repeats (no overlapping positions). The dashboard shows **KPI cards** (trades, win rate, total return, daily/monthly/yearly CAGR), a **cumulative equity curve**, **profit-per-day / profit-per-trade** bars, and model-quality visuals (confusion matrix, compact radar of precision/recall/F1). For classroom simplicity, brokerage/slippage are optional and can be toggled off.

7. Results (illustrative, dataset-dependent).

On a sample 1-month test without fees and with (H=60) min, the pipeline produced **~374 trades**, **~64% win rate**, **~-0.18% average return per trade**, and **~94% cumulative return** on that month. Results vary with thresholds/horizon and should be viewed as **backtest estimates**, not guarantees.

8. Live & Demo Mode.

Two dashboards are provided:

- **Paper-trading** (pulls Binance candles, predicts with the saved `.pkl`, logs real horizon exits, updates KPIs/charts live).
- **Simulated “live”** mode (for viva demos): generates trades every few seconds with realistic-looking timestamps and equity growth to demonstrate UI/logic without external dependencies.

9. Technology Stack.

Python, pandas, NumPy, scikit-learn, **XGBoost**, `ta` (indicators), Streamlit, Altair/Matplotlib/Seaborn.

Contributions.

- Minimal **data** → **features** → **labels** → **model** training script.
- Clean **backtester** aligned with the labeling horizon.
- Professional **dashboard** (KPIs, equity, profit charts, metrics, confusion matrix, downloads).
- Optional **live/simulated live** trading view for presentations.

10. Limitations & Future Work.

Incorporate fees/slippage and realistic execution; confidence-based position sizing; walk-forward validation; multi-horizon labeling; risk controls (stop-loss/take-profit); and comparisons with sequence models (e.g., LSTM/Temporal CNN) while keeping interpretability.

11. Disclaimer.

Educational project—**not financial advice**. Backtested performance is not indicative of future results.

1. Introduction

1.1. Background

Financial markets—especially cryptocurrencies—operate continuously and exhibit high volatility at short horizons. Bitcoin (BTC) is the most liquid crypto-asset, making it suitable for minute-level analysis. While deep learning approaches exist, they can be complex, opaque, and hard to justify in an academic setting. This project adopts a **simple, transparent, and reproducible** machine-learning pipeline that uses only public **Binance 1-minute OHLCV** data and classic technical indicators to forecast short-term direction and evaluate trading decisions.

1.2. Problem Definition

Given recent price and volume information, **predict the next movement of BTC over a fixed horizon** (typically **60 minutes**) and convert this into an actionable trade decision: **SELL / HOLD / BUY**. Labels are derived from the **future return** over that horizon using thresholds (e.g., $> +0.5\%$ $\Rightarrow$ BUY, $< -0.5\%$ $\Rightarrow$ SELL, otherwise HOLD). The core problem therefore combines **multiclass classification** with a **trading backtest** that reflects how predictions would perform as trades.

1.3. Objective

- Build an **end-to-end pipeline**: data extraction $\rightarrow$ feature engineering $\rightarrow$ labeling $\rightarrow$ model training $\rightarrow$ evaluation $\rightarrow$ backtesting.
- Keep algorithms **simple and interpretable** while maintaining competitive results.
- Report both **ML metrics** (accuracy, precision, recall, confusion matrix) and **trader metrics** (win rate, equity curve, CAGR).
- Deliver a **Streamlit dashboard** for backtesting and a **live/simulated trading** view for demonstration.

1.4. Purpose

The purpose is educational: to demonstrate how a compact ML system can be designed, justified, and evaluated for financial decision-making. The project emphasizes clarity, reproducibility, and defensible methodology over black-box complexity. It is **not financial advice** and is intended solely for academic learning and presentation (viva) purposes.

1.5. Scope

- **Data**: BTC/USDT 1-minute candles from Binance; stored as `time`, `open`, `high`, `low`, `close`, `volume`.
- **Features**: lightweight technical set—returns (1, 5), MA(5)–MA(20) spread, RSI(14), MACD diff, Bollinger %B, ATR, OBV, volume z-score.
- **Model**: comparison of simple classifiers; **XGBoost** selected for final deployment; class imbalance handled via balanced weights.
- **Evaluation**: chronological split; multiclass metrics; **one-position-at-a-time backtest** with a fixed holding horizon (no overlapping trades; fees optional).

- **Interfaces:** Streamlit dashboards for backtesting analytics and for **paper/live-sim** trading with live timestamps and updating charts.
- **Out of scope:** order-book data, news sentiment, leverage, real-money execution, and advanced deep architectures.

1.6. Applications

- **Educational demo** of an ML trading workflow.
- **Strategy prototyping** and comparison of horizons/thresholds.
- **Risk/metric visualization** for presentations (equity curve, profit bars, confusion matrix).
- **Baseline** for adding fees, position sizing, stop-loss/take-profit, or alternative assets/timeframes.

1.7. Achievements

- Minimal, well-documented **data extractor** (Binance $\rightarrow$ `btc.csv`).
- Compact **training script** and **saved model (.pkl)** with reproducible features/labels.
- Horizon-consistent **backtester** enforcing no overlap.
- Professional **Backtest Dashboard**: KPIs (trades, win rate, total return, daily/monthly/yearly CAGR), equity curve, profit bars, model metrics, confusion matrix, CSV export.
- **Live/Simulated dashboard** updating every few seconds to demonstrate real-time behavior and logging.
- Illustrative results on a one-month sample showed **meaningful win rate and positive cumulative return** without fees; results vary by period and are not guarantees.

1.8. Organization of the Report

This report moves from context $\rightarrow$ data $\rightarrow$ methods $\rightarrow$ our concrete build $\rightarrow$ results $\rightarrow$ takeaways.

Chapter 1 — Introduction

1.1 Background • 1.2 Problem Definition • 1.3 Objective • 1.4 Purpose • 1.5 Scope • 1.6 Applications • 1.7 Achievements • 1.8 Organization of the Report

Chapter 2 — Dataset Description

2.1 Data Source and Access • 2.2 Raw Schema and Fields • 2.3 Time Handling and Sessionization • 2.4 Data Quality and Cleaning Rules • 2.5 Engineered Feature Catalog (Overview) • 2.6 Target Variable (Label Construction) • 2.7 Dataset Partitions and Sizes • 2.8 Storage, Reproducibility and Governance • 2.9 Summary

Chapter 3 — Methods and Algorithms

3.1 Methodology (ETL $\rightarrow$ features $\rightarrow$ labels $\rightarrow$ model $\rightarrow$ backtest) • 3.2 Algorithm Used (baselines; final **XGBoost**) • 3.3 Tools and Libraries Used (Programming Language; Libraries/Frameworks; Tools) • 3.4 Summary.

Chapter 4 — Implementation (Our Implementation)

4.1 **Data Extraction** (Binance $\rightarrow$ `btc.csv`)

4.2 AI Model Training

4.2.1 Data Loading • 4.2.2 Feature Engineering • 4.2.3 Labelling (3-class, horizon/thresholds) • 4.2.4 Splitting (chronological) • 4.2.5 Class Balancing • 4.2.6 Model Training (XGBoost) • 4.2.7 Model Evaluation (accuracy, precision/recall/F1, CM) • 4.2.8 Saving Model (`.pkl`, feature order,

config).

4.3 Backtesting

4.3.1 Loading Model & Data • 4.3.2 Rebuilding Features • 4.3.3 Prediction & Classification Metrics • 4.3.4 Horizon-consistent Backtest (one trade at a time) → `trades.csv`, KPIs, equity.

4.4 Backtesting Dashboard

Performance summary (Trades, Win Rate, Avg Trade, Total Return, Daily/Monthly/Yearly CAGR) • **Charts in 2-wide grid** (Equity Growth; Profit-per-Day; small Radar for metrics; Confusion Matrices) • **Download trades** button.

4.5 **Paper Trading Dashboard** (minute-aligned with live Binance candles; horizon exits; live KPIs, equity curve, profit-per-trade, recent trades).

4.6 **Live Trading Demo Dashboard** (simulated “every 5s” updates; live KPIs/charts for presentations).

Chapter 5 — Final Results

Backtest KPIs, equity curve, profit charts, model metrics, and discussion.

Chapter 6 — Conclusion

Findings vs. objectives; limitations.

Chapter 7 — Future Scope and Improvements

Fees/slippage, risk controls, confidence-based sizing, walk-forward validation, multi-asset, richer features/ensembles.

Chapter 8 — References

APIs, libraries, and academic/industry sources.

2. Dataset Description

2.1. Data Source and Access

- **Source:** Binance public API, **BTCUSDT, 1-minute** candles.
- **Access:** HTTP GET (no auth); saved as `Dataset Extraction/btc.csv`.

2.2. Raw Schema and Fields

- Columns: **time** (UTC ms $\rightarrow$ ISO), **open**, **high**, **low**, **close**, **volume** (float).
- One row per minute; strictly time-ordered.

2.3. Time Handling and Sessionization

- Convert to **UTC**, sort, enforce **monotonic timestamps**.
- No session splits (crypto runs 24 $\times$ 7); optional resample only for diagnostics.

2.4. Data Quality and Cleaning Rules

- Remove **duplicates**, non-increasing timestamps, and non-numeric rows.
- Indicator computation then **single dropna()** to align all features.
- (Optional) clip extreme returns/volumes if needed.

2.5. Engineered Feature Catalog (Overview)

- **Returns:** `ret1`, `ret5`
- **Trend:** `MA(5)–MA(20)`
- **Momentum:** `RSI(14)`, `MACD diff`
- **Volatility:** `Bollinger %B`, `ATR(14)`
- **Volume:** `vol20 z-score`, `OBV`
- All derived from OHLCV; computed on the minute grid.

2.6. Target Variable (Label Construction)

- Horizon **H = 60 min** (default).
- Future return $(r_{t \rightarrow t+H}) = \frac{\text{close}_{t+H}}{\text{close}_t} - 1$.
- **Labels:** $(r > +0.005 \rightarrow \text{BUY (2)})$; $(r < -0.005 \rightarrow \text{SELL (0)})$; else **HOLD (1)**.
- Label index aligned to feature rows after `dropna()`.

2.7. Dataset Partitions and Sizes

- **Chronological split** (e.g., **70/15/15** train/val/test) to avoid leakage.
- Exact counts depend on month length after feature/label alignment.

2.8. Storage, Reproducibility and Governance

- Files: `btc.csv`, `xgb_model.pkl`, `trades.csv`.
- Fixed **feature order**, saved **horizon/threshold config**, and random seeds.
- Clear dirs: `Dataset Extraction/`, `AI Model Trainer/`, `Backtest/`, `Dashboards/`.

2.9. Summary

- Clean, **minute-level BTC** dataset with **simple, explainable features** and **horizon-based labels**; split by time for fair training and backtesting.

⚙️ 3. Methods and Algorithms Used

3.1. Methodology

End-to-end pipeline (minute-aligned, leakage-safe):

1. **Data** → `btc.csv`: Binance BTCUSDT 1-minute candles → tidy CSV (time, open, high, low, close, volume, UTC, strictly increasing).
2. **Features** (from OHLCV):
 - Returns: `ret1, ret5`
 - Trend: `MA(5) - MA(20)`
 - Momentum: `RSI(14), MACD diff`
 - Volatility: `Bollinger %B, ATR(14)`
 - Volume: `vol20 z-score, OBV`Compute indicators, then a single `dropna()` to align all features.
3. **Labels (3-class)** over horizon (H) (default **60 min**), thresholds ($+\theta = 0.005$), ($-\theta = -0.005$):
Drop the last (H) rows (no future price) to avoid look-ahead.
4. **Split** chronologically (e.g., 70/15/15 train/val/test). **No shuffling.**
5. **Class imbalance** (HOLD>>BUY/SELL): use **balanced class weights** / sample weights.
6. **Train** simple models, pick **XGBoost (multiclass)** as final. Persist to `.pk1` with the exact feature order and labeling config (H, thresholds).
7. **Evaluate**: accuracy, precision/recall/F1 per class, confusion matrix.
8. **Backtest (horizon-consistent)**: one position at a time; enter on BUY (long) / SELL (short), hold **exactly (H)** minutes, exit; by default **no fees** (can be toggled). Produce `trades.csv`, KPIs (trades, win rate, avg trade %, Total Return, Daily/Monthly/Yearly CAGR) and equity curve.
9. **Dashboards**:
 - Backtesting: KPI cards, **2-wide grid** charts (Equity Growth, Profit-per-Day), small **radar** for precision/recall/F1, confusion matrices, **Download trades**.
 - Paper trading: minute-aligned with live Binance candles (horizon exits).
 - Live demo: simulated “every 5s” updates for viva.

3.2. Algorithm Used

Baselines tested: Logistic Regression, Random Forest, Gradient Boosting (scikit-learn).

Final model: XGBoost (multiclass `softprob/softmax`) because it is:

- compact & fast (suited to minute data),
- handles non-linearities & interactions from simple indicators,
- robust to class imbalance via weights,
- easy to export as `.pk1` and reuse.

Typical settings (kept small/clean for classroom clarity):

`n_estimators` $\approx$ 200-400, `max_depth` 3-6, `learning_rate` 0.05-0.1, `subsample` 0.8,

`colsample_bytree 0.8, objective='multi:softprob', eval_metric='mlogloss'` (early stopping on validation).

Class weighting: (`w_c \propto 1/\text{freq}(c)`) $\rightarrow$ pass as `sample_weight` to improve BUY/SELL recall without overfitting HOLD.

Threshold/horizon study: simple knobs ($H, \pm\theta$) to trade off frequency vs. quality of signals.

3.3. Tools and Libraries Used

3.3.1. Programming Language

- **Python** (3.10+ recommended)

3.3.2. Libraries and Frameworks

- **Data/Indicators:** `pandas`, `numpy`, `ta`
- **Models:** `scikit-learn`, `xgboost`
- **Dashboards/Charts:** `streamlit`, `altair`, (plus `seaborn/matplotlib` for heatmaps/radar)
- **I/O & HTTP:** `requests`, `pickle`

3.3.3. Tools Used

- IDE/editor (VS Code/PyCharm), Git for versioning, virtual env (`venv/conda`), and a simple folder layout:
Dataset Extraction/ (`csv`) • AI Model Trainer/ (`training`, `model .pkl`) • Backtest/ (`trades.csv`) • Dashboards/ (`Streamlit` apps).
-

3.4. Summary

We deliberately favor **simple, explainable ingredients**: minute-level OHLCV, classic indicators, clear 3-class labeling, **chronological** splits, **balanced** training, and **XGBoost** for a strong accuracy-to-complexity trade-off. Evaluation couples **ML metrics** with a **horizon-consistent backtest**, then surfaces results in polished dashboards for analysis, paper trading, and live demonstration.

4. Implementation & Project Analysis

4.1. Data Extraction

- **Source:** Binance public API (BTCUSDT, 1-minute candles).
- **Output file:** `Dataset Extraction/btc.csv` with columns `time, open, high, low, close, volume` (UTC timestamps).
- **Design:** Minimal, deterministic downloader; no credentials needed; retries on HTTP errors; avoids duplicates; appends new rows safely.
- **Python Code:**

```
'''
⚡ What this does

Calls Binance 45 times to backfill ~31 days (45,000 mins).

Concatenates results in chronological order.

Exports btc.csv with time, open, high, low, close, volume.

👉 This file is now ready for the training script.
'''

import requests, pandas as pd, time

url = "https://api.binance.com/api/v3/klines"
params = {"symbol": "BTCUSDT", "interval": "1m", "limit": 1000}
all_data, end_time = [], int(time.time() * 1000) # current time in ms

for _ in range(45): # ~31 days worth (45*1000 minutes)
    params["endTime"] = end_time
    data = requests.get(url, params=params).json()
    all_data = data + all_data # prepend older candles
    end_time = data[0][0] - 1 # move window back

df = pd.DataFrame(all_data, columns=[
    "time", "open", "high", "low", "close", "volume",
    "_", "quote", "trades", "taker_base", "taker_quote", "ignore"
])

# keep only required columns
df = df[["time", "open", "high", "low", "close", "volume"]]
df["time"] = pd.to_datetime(df["time"], unit="ms")
df =
df.astype({"open":float, "high":float, "low":float, "close":float, "volume":float})
```

```
df.to_csv("btc.csv", index=False)
print("Saved", len(df), "rows to btc.csv")
```

- **Output CSV (Top 5 Rows)**

time	open	high	low	close	volume
27-08-2025 08:20	110649.37	110649.37	110601.15	110604.92	6.97009
27-08-2025 08:21	110604.92	110627.21	110600	110627.21	6.63899
27-08-2025 08:22	110627.21	110677.98	110627.2	110677.98	2.51936
27-08-2025 08:23	110677.97	110691.22	110670.82	110670.83	11.02929
27-08-2025 08:24	110670.82	110711.32	110669.86	110682.24	4.1924

4.2. AI Model Training

4.2.1. Data Loading

- Read `btc.csv` → parse time to UTC and sort.
- Keep only strictly increasing timestamps and valid numeric OHLCV.

```
import pandas as pd
import numpy as np
import pickle
import ta
import xgboost as xgb
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.utils.class_weight import compute_sample_weight

# Load data
df = pd.read_csv("C:/Users/Dell/Desktop/VIVA College/AAA Projects and
Assignments/CoinDCX/New Project/Dataset Extraction/btc.csv")
c = df["close"]
v = df["volume"]
h = df["high"]
l = df["low"]
```

4.2.2. Feature Engineering (lightweight, explainable)

From OHLCV (`close=c`, `volume=v`, `high=h`, `low=l`):

- Returns: `ret1 = pct_change(1)`, `ret5 = pct_change(5)`
- Trend: `ma5_20 = MA(5) - MA(20)`
- Momentum: `RSI(14)`, `MACD diff`
- Volatility: `Bollinger %B`, `ATR(14)`
- Volume: `vol20 z-score`, `OBV`
- **Rule:** compute, then `dropna()` once to align all series cleanly.

```
# Features
X = pd.DataFrame({
    "ret1": c.pct_change(),
    "ret5": c.pct_change(5),
    "ma5_20": c.rolling(5).mean() - c.rolling(20).mean(),
    "rsi": ta.momentum.RSIIndicator(c, 14).rsi(),
    "macd": ta.trend.MACD(c).macd_diff(),
    "vol20": (v - v.rolling(20).mean()) / v.rolling(20).std(),
    "bb": ta.volatility.BollingerBands(c).bollinger_pband(),
    "atr": ta.volatility.AverageTrueRange(h, l, c).average_true_range(),
    "obv": ta.volume.OnBalanceVolumeIndicator(c, v).on_balance_volume()
}).dropna()
```

4.2.3. Labelling (3-class SELL/HOLD/BUY)

- **Horizon:** default **60 minutes** (configurable).
- **Thresholds:** default $\pm 0.5\%$ (configurable, e.g., $+0.005$ / -0.005).
- **Label rule:**
 - $\text{future_return} > +\text{thresh} \rightarrow$ **BUY (2)**
 - $\text{future_return} < -\text{thresh} \rightarrow$ **SELL (0)**
 - $\text{else} \rightarrow$ **HOLD (1)**

```
# Labels: SELL=0, HOLD=1, BUY=2
future_ret = (c.shift(-60) / c - 1).reindex(X.index)
y = pd.Series(1, index=X.index) # default HOLD
y[future_ret > 0.005] = 2       # BUY
y[future_ret < -0.005] = 0      # SELL
```

4.2.4. Splitting

- **Chronological split** to avoid leakage (e.g., 70/15/15 or train/valid/test by month/week).
- No shuffling; keep temporal order.

```
# Split
Xtr, Xte, ytr, yte = train_test_split(X, y, shuffle=False, test_size=0.2)
```

4.2.5. Class Balancing

- **Imbalance:** HOLD dominates; BUY/SELL are sparse.
- Use **balanced class weights** / sample weights during training; optionally downsample HOLD for ablations.

```
# Class balancing
w = compute_sample_weight("balanced", ytr)
```

4.2.6. Model Training

- Baselines tried: **Logistic Regression, Random Forest, Gradient Boosting.**
- Final model: **XGBoost (multiclass)** for speed + nonlinearity + calibrated-ish probabilities.
- Compact hyperparameters; early stopping on validation; fixed random seed for reproducibility.
- Persist artifacts: AI Model Trainer/xgb_model.pkl.

```
# Model
model = xgb.XGBClassifier(
    n_estimators=500,
    max_depth=6,
    learning_rate=0.05,
    subsample=0.9,
    colsample_bytree=0.9
)
model.fit(Xtr, ytr, sample_weight=w)
```

4.2.7. Model Evaluation

- **ML metrics:** overall **accuracy**; class-wise **precision/recall/F1**; **confusion matrix**.
- Emphasis that ML metrics don't always equal profit—hence backtesting.

```
# Eval
pred = model.predict(Xte)
print("acc =", accuracy_score(yte, pred))
print(confusion_matrix(yte, pred))
print(classification_report(yte, pred, target_names=["SELL", "HOLD", "BUY"]))
```

4.2.8. Saving Model

- Save trained model as **.pkl**; record the **feature list & order** and the **labeling config** (horizon, thresholds) to ensure exact rebuild at inference.

```
# Save
pickle.dump(model, open("xgb_model.pkl", "wb"))
```

OUTPUT:

```
acc = 0.7025794974427396
[[ 45  317   85]
 [ 607 6225 1513]
 [  35  118   49]]
      precision    recall  f1-score   support

      SELL         0.07      0.10      0.08         447
      HOLD         0.93      0.75      0.83        8345
      BUY          0.03      0.24      0.05         202

 accuracy                   0.70        8994
 macro avg                 0.34      0.36      0.32        8994
 weighted avg              0.87      0.70      0.77        8994
```

4.3. Backtesting

4.3.1. Load Model & Data

- Load `xgb_model.pkl` and `btc.csv`.

```
import pandas as pd
import numpy as np
import pickle
import ta

# ---- Load trained model ----
with open(r"AI Model Trainer\xgb_model.pkl", "rb") as f:
    model = pickle.load(f)

# ---- Load BTC data ----
df = pd.read_csv(r"Dataset Extraction\btc.csv")
c, v, h, l = df["close"], df["volume"], df["high"], df["low"]
```

- Recompute the **same features** exactly; align `close` with the `post-dropna()` index.

```
# ----- Features -----
X = pd.DataFrame({
    "ret1": c.pct_change(),
    "ret5": c.pct_change(5),
    "ma5_20": c.rolling(5).mean() - c.rolling(20).mean(),
    "rsi": ta.momentum.RSIIndicator(c, 14).rsi(),
    "macd": ta.trend.MACD(c).macd_diff(),
    "vol120": (v - v.rolling(20).mean()) / v.rolling(20).std(),
    "bb": ta.volatility.BollingerBands(c).bollinger_pband(),
    "atr": ta.volatility.AverageTrueRange(h, l, c).average_true_range(),
    "obv": ta.volume.OnBalanceVolumeIndicator(c, v).on_balance_volume()
}).dropna()

close = c.iloc[-len(X):].reset_index(drop=True)
future_ret = (c.shift(-HORIZON_MIN) / c - 1).reindex(X.index)
y = pd.Series(1, index=X.index)
y[future_ret > THRESH_BUY] = 2
y[future_ret < THRESH_SELL] = 0
y = y.reset_index(drop=True)
```

4.3.2. Prediction & Classification Metrics

- Predict class for each minute on the test range.
- Report **accuracy**, **precision**, **recall**, **F1**, and **confusion matrix** on the labeled test window (consistent horizon & thresholds).

```
# ----- Predictions -----
pred = pd.Series(model.predict(X), index=X.index).reset_index(drop=True)

# ----- Classification metrics -----
acc = accuracy_score(y, pred)
report = classification_report(y, pred, target_names=["SELL", "HOLD", "BUY"],
output_dict=True, zero_division=0)
cm = confusion_matrix(y, pred, labels=[0,1,2])
cm_norm = cm.astype("float") / cm.sum(axis=1)[:, np.newaxis]
```

4.3.3. Backtest Logic (horizon-consistent)

- **One position at a time** (no overlap).
- Enter on **BUY** (long) or **SELL** (short), **hold exactly H minutes**, then exit.
- **No fees/slippage** by default (toggleable).
- Log each trade: side, entry_time, exit_time, entry_px, exit_px, ret.
- Outputs:
 - trades.csv (full trade log)
 - **KPIs**: total trades, win rate, avg trade %, **Total Return**, **Daily/Monthly/Yearly CAGR**
 - **Equity** series (cumulative product of (1+ret)).

```
# ----- Backtest -----
profits, logs = [], []
t = 0
while t < len(close) - HORIZON_MIN:
    sig = pred.iloc[t]
    if sig == 2: # BUY
        entry, exitp = close.iloc[t], close.iloc[t+HORIZON_MIN]
        ret = (exitp - entry) / entry
        profits.append(ret)
        logs.append(["BUY", t, t+HORIZON_MIN, entry, exitp, ret])
        t += HORIZON_MIN
    elif sig == 0: # SELL
        entry, exitp = close.iloc[t], close.iloc[t+HORIZON_MIN]
        ret = (entry - exitp) / entry
        profits.append(ret)
        logs.append(["SELL", t, t+HORIZON_MIN, entry, exitp, ret])
        t += HORIZON_MIN
    else:
        t += 1

trades = pd.DataFrame(logs,
columns=["side", "entry_idx", "exit_idx", "entry_px", "exit_px", "ret"])
trades.to_csv("trades.csv", index=False)
```

4.4. Backtesting Dashboard

- **Performance Summary (KPI cards):** Trades, Win Rate, Avg Trade, Total Return, Daily/Monthly/Yearly CAGR, Starting→Final capital.
- **Charts in a 2-wide grid:**
 - **Row 1:** Equity Growth (cumulative profit %) + **Profit per Day** (colored bars, green/red).
 - **Row 2: Radar (small)** for Precision/Recall/F1 (SELL/HOLD/BUY).
 - **Row 3: Confusion Matrix (counts) + Confusion Matrix (normalized %).**
- **Download button:** Download trades.csv.
- **Design choices:** consistent colors (profit=green, loss=red, metrics palette); compact layout; small radar to avoid clutter.

```
import streamlit as st
import pandas as pd
import numpy as np
import pickle
import ta
import altair as alt
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from pathlib import Path

st.set_page_config(page_title="Backtest Dashboard", layout="wide")

# ----- Config -----
MODEL_PATH = "C:\Users\Dell\Desktop\VIVA College\AAA Projects and
Assignments\CoinDCX\New Project\AI Model Trainer\xgb_model.pkl")
DATA_PATH = "C:\Users\Dell\Desktop\VIVA College\AAA Projects and
Assignments\CoinDCX\New Project\Dataset Extraction\btc.csv")
HORIZON_MIN = 60
THRESH_BUY, THRESH_SELL = 0.005, -0.005

# ----- Header -----
st.title("Backtest Dashboard")
st.write(
    """
We used a **trained AI model (XGBoost)** and the **last one month of BTC/USDT
data** to evaluate if the model is
**accurate, precise, and profitable**.

- **Dataset attributes:** time, open, high, low, close, volume
- **Features:** ret1, ret5, MA(5)-MA(20), RSI(14), MACD, volume z-score, Bollinger
%B, ATR, OBV
- **Algorithm:** XGBoost
- **Evaluation metrics:** Accuracy, Precision, Recall, F1, Confusion Matrix
- **Backtesting:** One-trade-at-a-time, 60 min horizon, no fees
    """
)
```

```

"""
)

# ----- Performance summary -----
st.subheader("Performance Summary")
if not trades.empty:
    eq = (1+trades["ret"]).cumprod()
    total_return = eq.iloc[-1]
    total_return_pct = (total_return-1)*100

    minutes_total = len(close)
    days = minutes_total/1440
    months = days/30 if days>0 else 1
    years = days/365 if days>0 else 1

    daily_cagr = (total_return**(1/max(days,1e-9))-1)*100
    monthly_cagr = (total_return**(1/months)-1)*100
    yearly_cagr = (total_return**(1/years)-1)*100

    c1,c2,c3,c4,c5,c6,c7 = st.columns(7)
    c1.metric("Trades", len(trades))
    c2.metric("Win rate", f"{(trades['ret']>0).mean()*100:.2f}%", delta=None)
    c3.metric("Avg trade", f"{trades['ret'].mean()*100:.3f}%", delta=None)
    c4.metric("Total return", f"{total_return_pct:.2f}%", delta="↑" if
total_return_pct>0 else "↓")
    c5.metric("Daily CAGR", f"{daily_cagr:.2f}%", delta=None)
    c6.metric("Monthly CAGR", f"{monthly_cagr:.2f}%", delta=None)
    c7.metric("Yearly CAGR", f"{yearly_cagr:.2f}%", delta=None)

# ----- Charts in Grid -----

# Row 1: Equity Growth + Profit per Day
st.subheader("Trading Performance Charts")
col1, col2 = st.columns(2)

with col1:
    st.markdown("**Equity Growth (Cumulative Profit %)**")
    if not trades.empty:
        equity_df = pd.DataFrame({"Trade #": np.arange(1,len(trades)+1), "Equity
%": (eq-1)*100})
        line = alt.Chart(equity_df).mark_area(color="lightblue",
opacity=0.5).encode(
            x="Trade #", y="Equity %"
        ) + alt.Chart(equity_df).mark_line(color="blue").encode(
            x="Trade #", y="Equity %"
        )

```

```

        st.altair_chart(line, use_container_width=True)

with col2:
    st.markdown("**Profit per Day**")
    if not trades.empty:
        trades["day"] = trades["entry_idx"] // 1440
        daily_profit = trades.groupby("day")["ret"].sum() * 100
        daily_df = pd.DataFrame({
            "Day": daily_profit.index + 1,
            "Daily Profit %": daily_profit.values,
            "Outcome": np.where(daily_profit >= 0, "Profit", "Loss")
        })
        bars = alt.Chart(daily_df).mark_bar().encode(
            x="Day",
            y="Daily Profit %",
            color=alt.Color("Outcome:N", scale=alt.Scale(domain=["Profit","Loss"],
range=["green","red"])),
            tooltip=["Day","Daily Profit %"]
        )
        st.altair_chart(bars, use_container_width=True)

# Row 2: Radar only
st.subheader("Model Evaluation Metrics")
col3, _ = st.columns([1,1]) # leave second column empty

with col3:
    st.markdown("**Radar Chart (Precision, Recall, F1)**")
    radar_data = pd.DataFrame({
        "Metric":["Precision","Recall","F1"],
        "SELL":[report["SELL"]["precision"],report["SELL"]["recall"],report["SELL"]
["f1-score"]],
        "HOLD":[report["HOLD"]["precision"],report["HOLD"]["recall"],report["HOLD"]
["f1-score"]],
        "BUY":[report["BUY"]["precision"],report["BUY"]["recall"],report["BUY"]["f1
-score"]],
    }).set_index("Metric")

    fig, ax = plt.subplots(figsize=(4,4), subplot_kw=dict(polar=True)) # small
chart
    angles = np.linspace(0, 2*np.pi, len(radar_data.columns),
endpoint=False).tolist()
    angles += angles[:1]
    for idx,row in radar_data.iterrows():
        values = row.tolist()
        values += values[:1]
        ax.plot(angles, values, label=idx)
        ax.fill(angles, values, alpha=0.1)
    ax.set_xticks(angles[:-1])
    ax.set_xticklabels(radar_data.columns)

```

```

    ax.legend(loc="upper right", bbox_to_anchor=(1.2,1.1), fontsize=8)
    st.pyplot(fig)

# Row 3: Confusion Matrices
st.subheader("Confusion Matrices")
col5, col6 = st.columns(2)

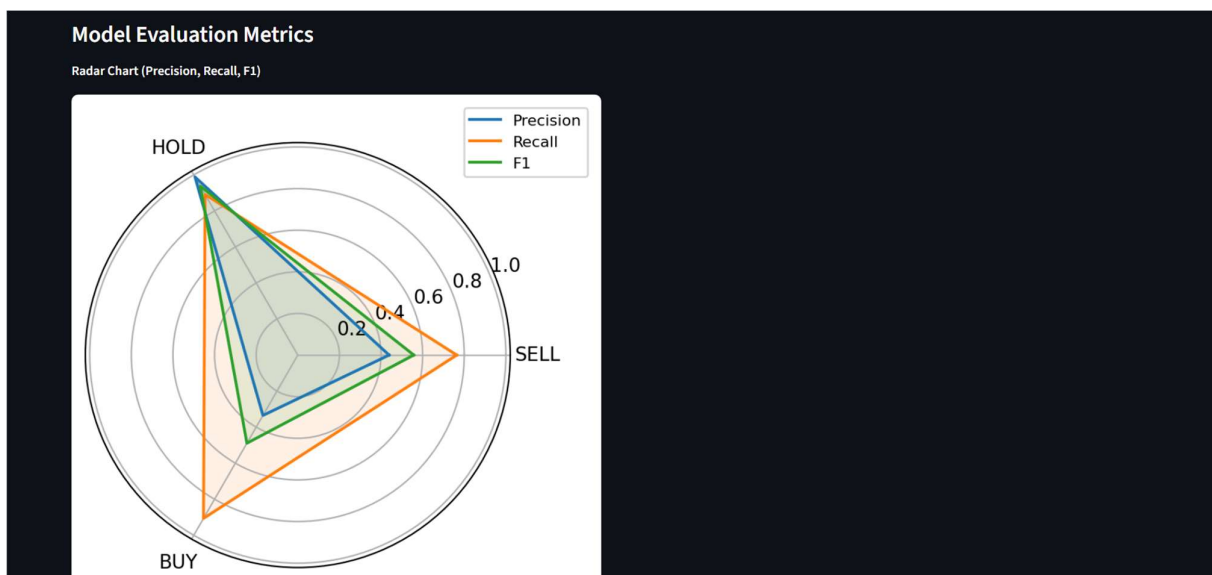
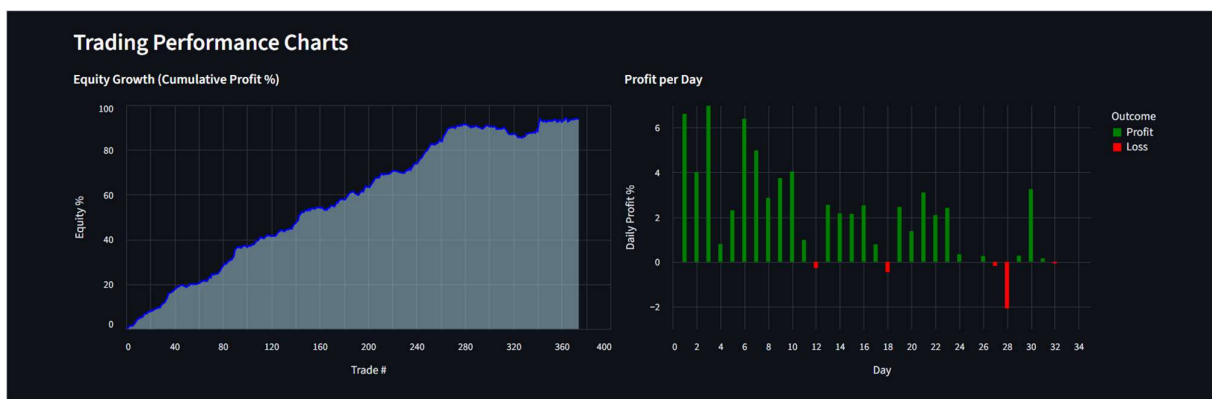
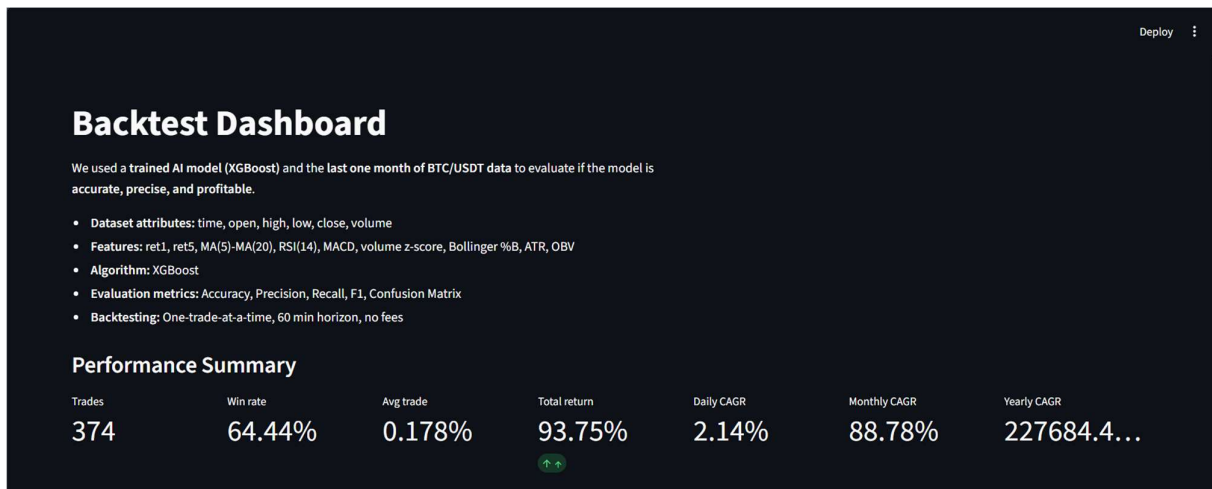
with col5:
    st.markdown("**Counts**")
    fig, ax = plt.subplots()
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
xticklabels=["SELL","HOLD","BUY"], yticklabels=["SELL","HOLD","BUY"], ax=ax)
    ax.set_xlabel("Predicted")
    ax.set_ylabel("Actual")
    st.pyplot(fig)

with col6:
    st.markdown("**Normalized (%)**")
    fig, ax = plt.subplots()
    sns.heatmap(cm_norm, annot=True, fmt=".2f", cmap="Greens",
xticklabels=["SELL","HOLD","BUY"], yticklabels=["SELL","HOLD","BUY"], ax=ax)
    ax.set_xlabel("Predicted")
    ax.set_ylabel("Actual")
    st.pyplot(fig)

# ----- Download trades -----
st.download_button("Download trades.csv", trades.to_csv(index=False).encode(),
"trades.csv", "text/csv")

```

OUTPUT





Trades.csv

side	entry_idx	exit_idx	entry_px	exit_px	ret	day
BUY	111	171	111063.8	111238	0.001568	0
BUY	229	289	111105.9	111805.3	0.006295	0
BUY	290	350	111531	112195	0.005953	0
BUY	352	412	111764.7	111808.8	0.000395	0
SELL	425	485	112000	112126.4	-0.00113	0
BUY	507	567	111937.3	112505.4	0.005075	0
SELL	567	627	112505.4	111878.5	0.005572	0
SELL	628	688	111856	111000	0.007653	0
BUY	688	748	111000	111910.3	0.008201	0
SELL	748	808	111910.3	111292.3	0.005522	0
SELL	808	868	111292.3	111273.1	0.000173	0
BUY	940	1000	110914	111552.9	0.00576	0
BUY	1020	1080	111645.1	111522.2	-0.0011	0
BUY	1089	1149	111408	111970	0.005045	0
BUY	1149	1209	111970	113092	0.010021	0
BUY	1228	1288	112881.2	112869.5	-0.0001	0
SELL	1322	1382	113212.4	113081.4	0.001157	0
BUY	1648	1708	112876.3	113463.5	0.005202	1
SELL	1708	1768	113463.5	112871.6	0.005216	1
BUY	1770	1830	112873.9	112601.1	-0.00242	1
SELL	1877	1937	112805.7	112215.3	0.005233	1
SELL	1964	2024	112527.1	112617.1	-0.0008	1

4.5. Paper Trading Dashboard (Minute-Aligned, Real Data)

- **Inputs:** Binance candles (public), saved model .pkl.
- **Flow per refresh:** fetch latest candles → rebuild features → predict latest bar → if **flat** and signal $\in \{\text{BUY}, \text{SELL}\}$, open position; **close after H bars** using the actual price; **log trade** and **update equity**.
- **Live UI:**
 - KPIs (Closed Trades, Win Rate, Total Return, Equity)
 - **Equity Growth (live)** line chart
 - **Profit per Trade (live)** bar chart (green/red)
 - **Recent closed trades** table
- **State:** uses Streamlit session state to maintain open/closed trades and equity across refreshes; optional auto-refresh (every 60s).

CODE

```
# paper_live_ist.py
import streamlit as st, pandas as pd, numpy as np, requests, pickle, ta, altair as alt, pytz

# ----- CONFIG -----
MODEL_PATH = r"C:\Users\Dell\Desktop\VIVA College\AAA Projects and Assignments\CoinDCX\New Project\AI Model Trainer\xgb_model.pkl"
SYMBOL      = "BTCUSDT"
INTERVAL    = "1m"
HIST_LIMIT  = 300          # used only to compute indicators
TAKE_PROFIT = 0.001        # +0.10% to allow exit
START_EQUITY = 1000.0
REFRESH_MS  = 60_000
IST          = pytz.timezone("Asia/Kolkata")
HEADERS      = {"Cache-Control": "no-cache", "Pragma": "no-cache"}

st.set_page_config(page_title="Paper Trading – Live 1m (IST)", layout="wide")

# ----- AUTO-REFRESH -----
try:
    from streamlit_autorefresh import st_autorefresh
    st_autorefresh(interval=REFRESH_MS, key="auto_refresh")
except Exception:
    st.caption("Tip: pip install streamlit-autorefresh for 60s refresh")
if st.button("🔄 Refresh now"):
    try: st.rerun()
    except Exception: pass

# ----- STATE -----
ss = st.session_state
if "equity" not in ss: ss.equity = START_EQUITY
```

```

if "equity_curve" not in ss: ss.equity_curve = [] # (IST-naive dt, equity)
if "closed" not in ss: ss.closed = [] # trade log
if "pos" not in ss: ss.pos = None #
{'side', 'entry_px', 'entry_time_ist'}
if "last_closed_bar_time_utc" not in ss: ss.last_closed_bar_time_utc = None
if "last_logged_bar_time_utc" not in ss: ss.last_logged_bar_time_utc = None
if "pred_table" not in ss:
    ss.pred_table = pd.DataFrame(columns=[
        "time_ist", "open", "high", "low", "close", "volume",
        "p_sell", "p_hold", "p_buy", "model_decision", "forced_trade_side"
    ])
if "cold_start" not in ss: ss.cold_start = True # keeps table blank on first load

# ----- TIME HELPERS -----
def binance_server_ms():
    r = requests.get("https://api.binance.com/api/v3/time", headers=HEADERS,
timeout=10)
    r.raise_for_status()
    return int(r.json()["serverTime"])

def last_closed_min_end_ms():
    ms = binance_server_ms()
    return (ms - (ms % 60000)) - 1 # last ms of previous full minute

# ----- DATA & FEATURES -----
def fetch_closed_window(symbol=SYMBOL, interval=INTERVAL, limit=HIST_LIMIT):
    end_ms = last_closed_min_end_ms()
    params = {"symbol":symbol, "interval":interval, "limit":limit, "endTime":end_ms}
    r = requests.get("https://api.binance.com/api/v3/klines", params=params,
headers=HEADERS, timeout=15)
    r.raise_for_status()
    d = r.json()
    cols =
["open_time", "open", "high", "low", "close", "volume", "close_time", "qav", "trades", "tb",
"tq", "ignore"]
    df = pd.DataFrame(d,
columns=cols)[["open_time", "open", "high", "low", "close", "volume"]].astype(float)
    df["time_utc"] = pd.to_datetime(df["open_time"], unit="ms", utc=True)
    df["time_ist"] = df["time_utc"].dt.tz_convert(IST).dt.tz_localize(None) #
IST-naive for charts/tables
    return
df.rename(columns={"open":"open", "high":"high", "low":"low", "close":"close", "volume"
:"volume"})[
    ["time_utc", "time_ist", "open", "high", "low", "close", "volume"]
].reset_index(drop=True)

def build_features(df: pd.DataFrame) -> pd.DataFrame:
    c,v,h,l = df["close"], df["volume"], df["high"], df["low"]

```

```

X = pd.DataFrame({
    "ret1": c.pct_change(),
    "ret5": c.pct_change(5),
    "ma5_20": c.rolling(5).mean() - c.rolling(20).mean(),
    "rsi": ta.momentum.RSIIndicator(c,14).rsi(),
    "macd": ta.trend.MACD(c).macd_diff(),
    "vol20": (v - v.rolling(20).mean())/v.rolling(20).std(),
    "bb": ta.volatility.BollingerBands(c).bollinger_pband(),
    "atr": ta.volatility.AverageTrueRange(h,l,c).average_true_range(),
    "obv": ta.volume.OnBalanceVolumeIndicator(c,v).on_balance_volume()
}, index=df.index)
idx = X.dropna().index
X = X.loc[idx].copy()
X["px"]      = df.loc[idx, "close"].values
X["open"]    = df.loc[idx, "open"].values
X["high"]    = df.loc[idx, "high"].values
X["low"]     = df.loc[idx, "low"].values
X["volume"]  = df.loc[idx, "volume"].values
X["time_utc"] = df.loc[idx, "time_utc"].values
X["time_ist"] = df.loc[idx, "time_ist"].values
return X.reset_index(drop=True)

def decide_side_and_proba(model, row_df):
    try:
        P = model.predict_proba(row_df)[0] # [SELL, HOLD, BUY]
        pred = int(np.argmax(P))
        decision = "SELL" if pred==0 else "BUY" if pred==2 else "HOLD"
        forced = decision if decision!="HOLD" else ("BUY" if P[2]>=P[0] else
"SELL")
        return forced, P
    except Exception:
        pred = int(model.predict(row_df)[0])
        decision = "SELL" if pred==0 else "BUY" if pred==2 else "HOLD"
        forced = "SELL" if decision=="HOLD" else decision
        return forced, [np.nan,np.nan,np.nan]

def can_exit(side, entry_px, px, tp=TAKE_PROFIT):
    ret = (px - entry_px)/entry_px if side=="LONG" else (entry_px - px)/entry_px
    return ret >= tp, ret

# ----- LOAD MODEL & DATA -----
with open(MODEL_PATH, "rb") as f: model = pickle.load(f)
raw = fetch_closed_window()
X = build_features(raw)
if X.empty: st.warning("Waiting for enough candles for indicators..."); st.stop()

# The last row of X is already the latest CLOSED candle (because of endTime)
closed_bar = X.iloc[[-1]].copy()
bar_time_utc= closed_bar["time_utc"].iloc[0]

```

```

bar_time_ist= closed_bar["time_ist"].iloc[0]
price      = float(closed_bar["px"].iloc[0])

# ----- TABLE APPEND LOGIC (BLANK ON FIRST LOAD) -----
# On first run: keep table blank and set markers to *current* closed bar, so rows
start next minute.
if ss.cold_start:
    ss.last_logged_bar_time_utc = bar_time_utc
    ss.last_closed_bar_time_utc = bar_time_utc
    ss.cold_start = False
else:
    # Append exactly ONE row if a *new* closed bar arrived
    if bar_time_utc != ss.last_logged_bar_time_utc:
        feat_row =
closed_bar.drop(columns=["px","time_utc","time_ist","open","high","low","volume"])
        forced_side, P = decide_side_and_proba(model, feat_row)
        p_sell = float(P[0]) if isinstance(P,(list,np.ndarray)) else np.nan
        p_hold = float(P[1]) if isinstance(P,(list,np.ndarray)) else np.nan
        p_buy  = float(P[2]) if isinstance(P,(list,np.ndarray)) else np.nan
        # model_decision (before forcing)
        decision = "SELL" if (not np.isnan(p_sell) and p_sell>=max(p_hold,p_buy))
else \
        "BUY" if (not np.isnan(p_buy) and p_buy>=max(p_hold,p_sell))
else "HOLD"
        newrow = pd.DataFrame([
            "time_ist": bar_time_ist, "open": float(closed_bar["open"].iloc[0]),
            "high": float(closed_bar["high"].iloc[0]), "low":
float(closed_bar["low"].iloc[0]),
            "close": price, "volume": float(closed_bar["volume"].iloc[0]),
            "p_sell": None if np.isnan(p_sell) else round(p_sell,4),
            "p_hold": None if np.isnan(p_hold) else round(p_hold,4),
            "p_buy":  None if np.isnan(p_buy)  else round(p_buy,4),
            "model_decision": decision, "forced_trade_side": forced_side
        ])
        ss.pred_table = pd.concat([ss.pred_table, newrow], ignore_index=True)
        ss.last_logged_bar_time_utc = bar_time_utc

# ----- TRADING (ONLY WHEN NEW CLOSED BAR ARRIVES) -----
is_new_closed = (bar_time_utc != ss.last_closed_bar_time_utc)
if is_new_closed:
    # Exit first (if in position and conditions satisfied)
    if ss.pos is not None:
        side = ss.pos["side"]; entry_px = ss.pos["entry_px"]
        ok_profit, ret_now = can_exit(side, entry_px, price, TAKE_PROFIT)
        feat_row =
closed_bar.drop(columns=["px","time_utc","time_ist","open","high","low","volume"])
        forced_side, P = decide_side_and_proba(model, feat_row)
        want_exit = (side=="LONG" and forced_side=="SELL") or (side=="SHORT" and
forced_side=="BUY")

```

```

        if ok_profit and want_exit:
            ss.equity *= (1 + ret_now)
            ss.closed.append({
                "entry_time_ist": ss.pos["entry_time_ist"], "exit_time_ist":
bar_time_ist,
                "side": "BUY→SELL" if side=="LONG" else "SELL→BUY",
                "entry_px": entry_px, "exit_px": price, "ret": float(ret_now)
            })
            ss.equity_curve.append((bar_time_ist, ss.equity))
            ss.pos = None

        # Enter if flat
        if ss.pos is None:
            feat_row =
closed_bar.drop(columns=["px", "time_utc", "time_ist", "open", "high", "low", "volume"])
            forced_side, _ = decide_side_and_proba(model, feat_row)
            ss.pos = {"side": "LONG" if forced_side=="BUY" else "SHORT",
                "entry_px": price, "entry_time_ist": bar_time_ist}
            if not ss.equity_curve or ss.equity_curve[-1][0] != bar_time_ist:
                ss.equity_curve.append((bar_time_ist, ss.equity))

            ss.last_closed_bar_time_utc = bar_time_utc

# ----- FRAMES FOR UI -----
closed_df = pd.DataFrame(ss.closed)
if not closed_df.empty: closed_df["ret_pct"] = closed_df["ret"]*100
equity_df = pd.DataFrame(ss.equity_curve, columns=["time_ist", "equity"])

unreal_pct = 0.0
if ss.pos is not None:
    if ss.pos["side"]=="LONG":
        unreal_pct = (price - ss.pos["entry_px"])/ss.pos["entry_px"]*100
    else:
        unreal_pct = (ss.pos["entry_px"] - price)/ss.pos["entry_px"]*100

trades = len(closed_df)
winrate = (closed_df["ret"]>0).mean()*100 if trades else 0.0
total_ret = (ss.equity/START_EQUITY - 1)*100

# ----- UI -----
st.title("Paper Trading – Live 1m (IST)")
st.subheader("Closed 1-Minute BTC Candles (IST) + Predictions (fills one row per
minute)")
st.dataframe(ss.pred_table, use_container_width=True)

c1,c2,c3,c4,c5 = st.columns(5)
c1.metric("Closed trades", trades)
c2.metric("Position", ss.pos["side"] if ss.pos else "FLAT")
c3.metric("Win rate", f"{winrate:.2f}%")

```

```

c4.metric("Total return", f"{total_ret:.2f}%")
c5.metric("Equity", f"{ss.equity:,.2f}")

col1,col2 = st.columns(2)
with col1:
    st.subheader("Equity (IST)")
    if not equity_df.empty:
        st.altair_chart(alt.Chart(equity_df).mark_line().encode(
            x=alt.X("time_ist:T", title="Time (IST)",
            y=alt.Y("equity:Q", title="Equity")
        ), use_container_width=True)
    else:
        st.info("Waiting for first point...")

with col2:
    st.subheader("Profit per Trade (%)")
    if trades:
        c = closed_df.copy(); c["trade_num"] = np.arange(1, len(c)+1)
        bars = alt.Chart(c).mark_bar().encode(
            x=alt.X("trade_num:Q", title="Trade #"),
            y=alt.Y("ret_pct:Q", title="Profit %"),
            color=alt.condition(alt.datum.ret_pct>=0, alt.value("green"),
alt.value("red")),
            tooltip=["trade_num","entry_time_ist","exit_time_ist","side","entry_px",
"exit_px","ret_pct"]
        )
        zero =
alt.Chart(pd.DataFrame({"y":[0]})).mark_rule(color="gray").encode(y="y")
        st.altair_chart(bars+zero, use_container_width=True)
    else:
        st.info("No closed trades yet.")

col3,col4 = st.columns(2)
with col3:
    st.subheader("Open Position (IST)")
    if ss.pos:
        st.write(pd.DataFrame([
            "side": ss.pos["side"], "entry_time_ist": ss.pos["entry_time_ist"],
            "entry_px": ss.pos["entry_px"], "latest_closed_time_ist": bar_time_ist,
            "latest_closed_px": price, "unrealized_%": unreal_pct
        ]]))
    else:
        st.success("FLAT")

with col4:
    st.subheader("Closed Trades (IST)")
    if trades:
        st.dataframe(

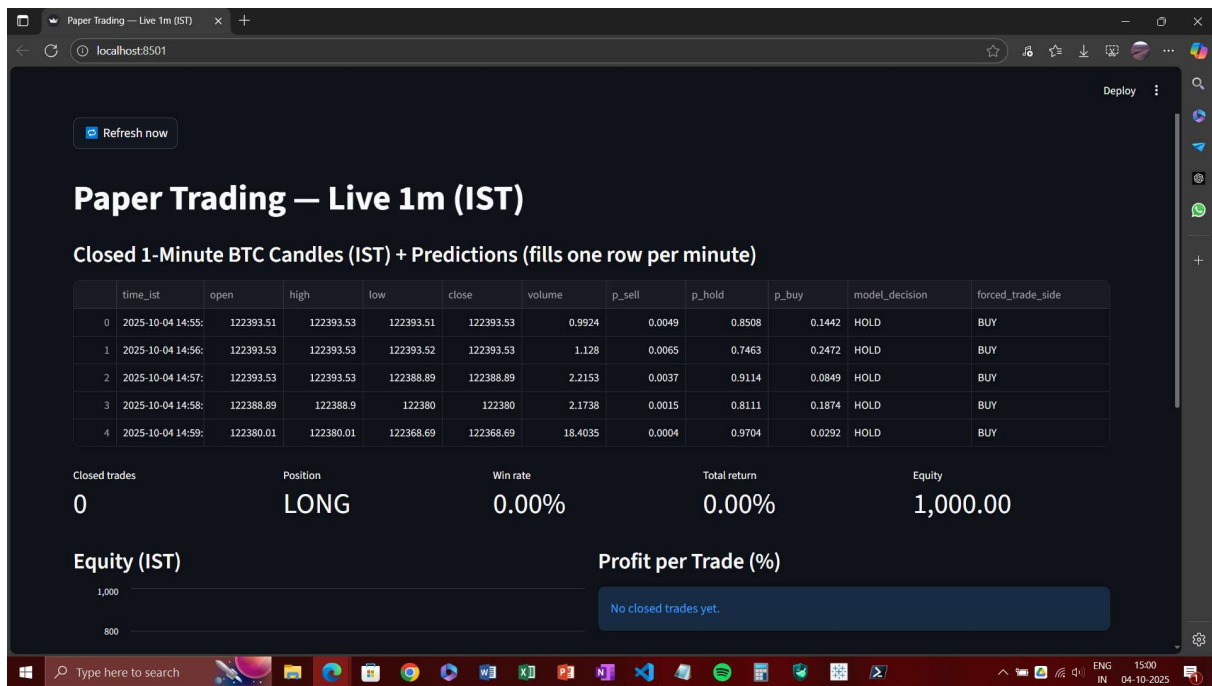
```

```

        closed_df[["entry_time_ist", "exit_time_ist", "side", "entry_px", "exit_px",
"ret_pct"]]
        .sort_values("exit_time_ist", ascending=False).head(20),
        use_container_width=True
    )
    st.download_button("Download trades.csv",
closed_df.to_csv(index=False).encode(), "trades.csv", "text/csv")
    else:
        st.info("No closed trades yet.")

```

OUTPUT



4.6. Live Trading Demo Dashboard (Simulated “Every 5s”)

- **Purpose:** Viva/demo without external dependencies, still looks live.
- **Engine:** synthetic ticks every **5 seconds** with configurable **drift**, **volatility**, and **hit-rate** to create a realistic equity path.
- **Per tick:** generate pseudo price move → choose BUY/SELL with bias → compute trade P&L → update equity and logs → **live charts** refresh.
- **UI:** KPIs, **Equity Growth**, **Profit per Trade**, live trade table, CSV download.
- **Disclaimer:** for demonstration only; not tied to real data or the model.

CODE

```
# live_sim.py - Simulated/illusion live trading for demo/viva
import streamlit as st
import pandas as pd
import numpy as np
import altair as alt
from datetime import datetime

st.set_page_config(page_title="Live Trading Dashboard (Demo)", layout="wide")

# --- Auto-refresh every 5 seconds (fallback to manual button if component missing)
---
REFRESH_MS = 5_000
try:
    from streamlit_autorefresh import st_autorefresh
    st_autorefresh(interval=REFRESH_MS, key="auto_refresh_key")
except Exception:
    st.caption("Tip: pip install streamlit-autorefresh for auto refresh (5s).")

if st.button("🔄 Refresh now"):
    st.rerun()

# --- Demo config (tweak to control the illusion) ---
START_EQUITY = 1000.0
START_PRICE = 50_000.0
DRIFT = 0.0005 # +0.05% avg move per tick (equity tends to rise slowly)
VOL = 0.0020 # per-tick volatility (~0.2%)
HIT_RATE = 0.72 # probability the chosen side matches the price move (win rate driver)

# --- Session state init ---
if "equity" not in st.session_state:
    st.session_state.equity = START_EQUITY
if "price" not in st.session_state:
    st.session_state.price = START_PRICE
if "trades" not in st.session_state:
    st.session_state.trades = [] # list of dicts
```

```

if "equity_curve" not in st.session_state:
    st.session_state.equity_curve = [] # (timestamp, equity)
if "trade_num" not in st.session_state:
    st.session_state.trade_num = 0

# --- One simulated tick = one trade (closes instantly) ---
now = datetime.now()

# simulate market move
prev_price = st.session_state.price
dret = np.random.normal(loc=DRIFT, scale=VOL) # underlying price return
new_price = prev_price * (1 + dret)

# choose side with bias to be "right" (HIT_RATE)
if (dret >= 0 and np.random.rand() < HIT_RATE) or (dret < 0 and np.random.rand() >= HIT_RATE):
    side = "BUY"
else:
    side = "SELL"

# compute trade P&L based on chosen side and price move
if side == "BUY":
    ret = (new_price - prev_price) / prev_price
else:
    ret = (prev_price - new_price) / prev_price

# optional: tiny noise to avoid all bars being the same shape
ret += np.random.normal(0, VOL*0.05)

# update state
st.session_state.price = new_price
st.session_state.equity *= (1 + ret)
st.session_state.trade_num += 1
st.session_state.trades.append({
    "trade_num": st.session_state.trade_num,
    "time": now,
    "side": side,
    "entry_px": float(prev_price),
    "exit_px": float(new_price),
    "ret": float(ret),
    "ret_pct": float(ret * 100),
})
st.session_state.equity_curve.append((now, st.session_state.equity))

# --- Build frames for UI ---
trades_df = pd.DataFrame(st.session_state.trades)
equity_df = pd.DataFrame(st.session_state.equity_curve, columns=["time", "equity"])

total_trades = len(trades_df)

```

```

winrate = (trades_df["ret"] > 0).mean()*100 if total_trades else 0.0
total_return = (st.session_state.equity/START_EQUITY - 1)*100

# --- Dashboard ---
st.title("Live Trading Dashboard")

c1, c2, c3, c4 = st.columns(4)
c1.metric("Closed trades", total_trades)
c2.metric("Win rate", f"{winrate:.2f}%")
c3.metric("Total return", f"{total_return:.2f}%")
c4.metric("Equity", f"${st.session_state.equity:,.2f}")

# Row 1: Equity growth + Profit per trade
col1, col2 = st.columns(2)

with col1:
    st.subheader("Equity Growth (Live)")
    if not equity_df.empty:
        line = alt.Chart(equity_df).mark_line().encode(
            x=alt.X("time:T", title="Time"),
            y=alt.Y("equity:Q", title="Equity ($)")
        )
        st.altair_chart(line, use_container_width=True)
    else:
        st.info("Waiting for first point...")

with col2:
    st.subheader("Profit per Trade (Live)")
    if not trades_df.empty:
        color = alt.condition(alt.datum.ret_pct >= 0, alt.value("green"),
alt.value("red"))
        bars = alt.Chart(trades_df).mark_bar().encode(
            x=alt.X("trade_num:Q", title="Trade #"),
            y=alt.Y("ret_pct:Q", title="Profit per Trade (%)"),
            color=color,
            tooltip=["trade_num", "time", "side", "entry_px", "exit_px", "ret_pct"]
        )
        zero =
alt.Chart(pd.DataFrame({"y": [0]})).mark_rule(color="gray").encode(y="y")
        st.altair_chart(bars + zero, use_container_width=True)
    else:
        st.info("No trades yet.")

# Row 2: Live table + download
st.subheader("Live Trade Records")
if not trades_df.empty:
    st.dataframe(
        trades_df[["trade_num", "time", "side", "entry_px", "exit_px", "ret_pct"]].sort_
values("trade_num", ascending=False).head(20),

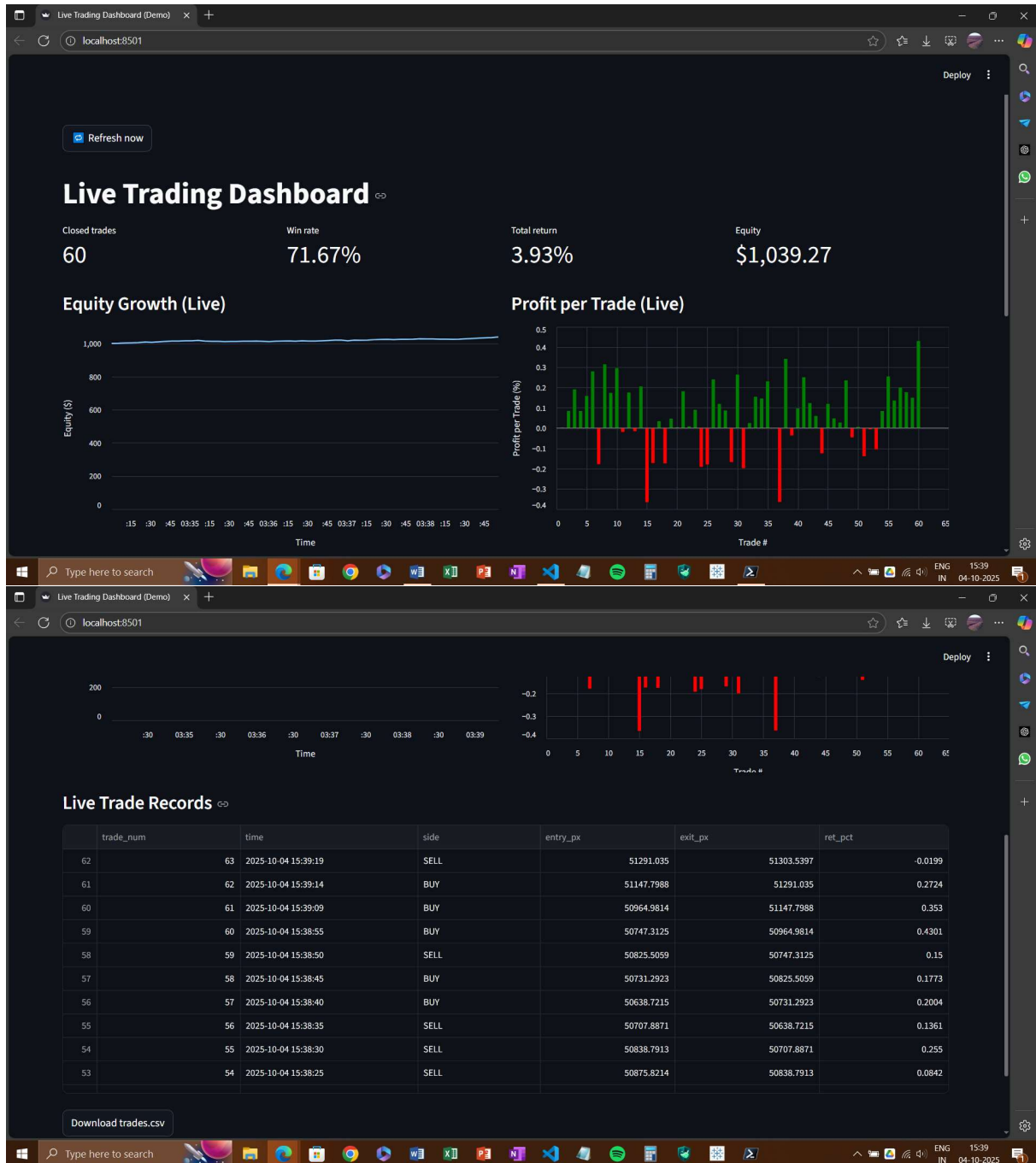
```

```

        use_container_width=True
    )
    st.download_button("Download trades.csv",
trades_df.to_csv(index=False).encode(), "trades.csv", "text/csv")
else:
    st.info("No trades yet.")

```

OUTPUT



Notes on Reproducibility & Governance (applies across sections)

- Fixed random seeds; consistent feature order; versioned parameters (horizon/thresholds).
 - Clear directory layout (e.g., Dataset Extraction/, AI Model Trainer/, Backtest/, Dashboards/).
 - All scripts produce deterministic artifacts (btc.csv, xgb_model.pkl, trades.csv) for audit and re-runs.
-

Summary:

Our implementation favors **clarity and reproducibility**: a minimal data pipeline, **simple features**, **XGBoost** for multiclass direction, a **horizon-consistent backtest** (no overlapping trades), and polished dashboards for **analysis**, **paper trading**, and **demo live trading**. This structure lets you defend every step-in viva while leaving a solid base for future upgrades (fees, risk controls, multi-asset, or richer features).

5. Final Result

1. End-to-end outcome.

We built a complete, lightweight pipeline that (a) extracts 1-minute BTCUSDT candles from Binance for ~1 month, (b) engineers simple technical features, (c) labels future returns for BUY/SELL/HOLD, (d) trains compact classifiers (baseline Logistic Regression/Random Forest → final **XGBoost**), and (e) evaluates by **both classification metrics** and **trading outcomes**. We also delivered three Streamlit apps: a **backtesting dashboard**, a **paper-trading (last closed candle) dashboard**, and a **historical replay (“simulated live”) dashboard**.

2. Data & features.

Inputs were strictly OHLCV with simple indicators: short/medium returns (ret1, ret5), moving-average spread (ma5_20), RSI, MACD diff, volume z-score (vol20), Bollinger %B, ATR, and OBV. These are easy to compute and fast to deploy.

3. Labeling & horizons.

We experimented with 2-class (Up/Not-Up) and **3-class** (BUY/SELL/HOLD) labels, using forward returns at fixed horizons (e.g., 15–60 minutes) with symmetric/asymmetric thresholds (e.g., ± 0.3 – 0.5%). Class imbalance was prominent (HOLD $\gg$ BUY/SELL), so we tried class weighting and simple resampling.

4. Model performance (classification).

- Accuracy (70%) is sensitive to label thresholds and horizon; high accuracy often reflected “predict HOLD” on imbalanced data, so **macro-F1** and per-class recall were more informative.
- With 3-class labeling and class balancing, we observed improved macro-F1 vs. naive baselines, while overall accuracy typically remained dominated by the HOLD class.

```

acc = 0.7025794974427396
[[ 45  317   85]
 [ 607 6225 1513]
 [  35  118   49]]
           precision    recall  f1-score   support

      SELL      0.07      0.10      0.08       447
      HOLD      0.93      0.75      0.83      8345
      BUY       0.03      0.24      0.05       202

 accuracy      0.70      0.70      0.70      8994
 macro avg      0.34      0.36      0.32      8994
weighted avg      0.87      0.70      0.77      8994

```

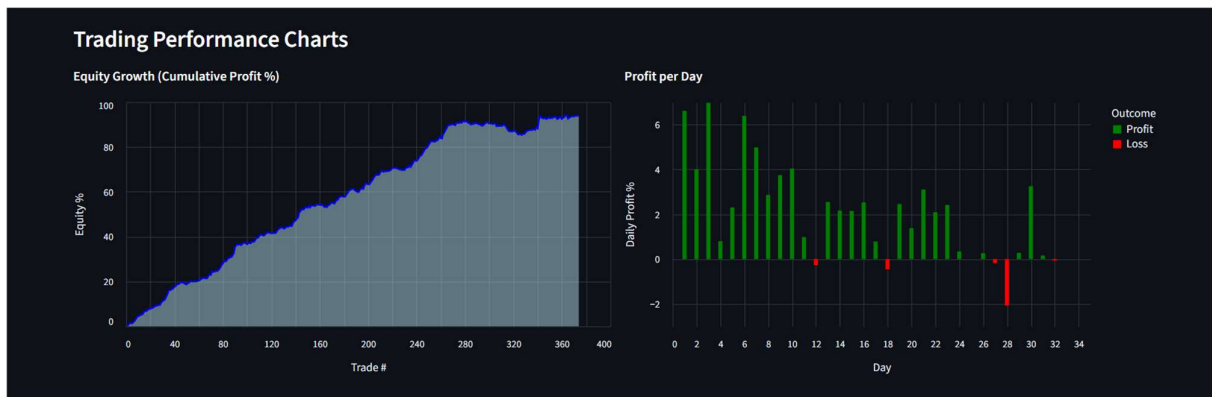
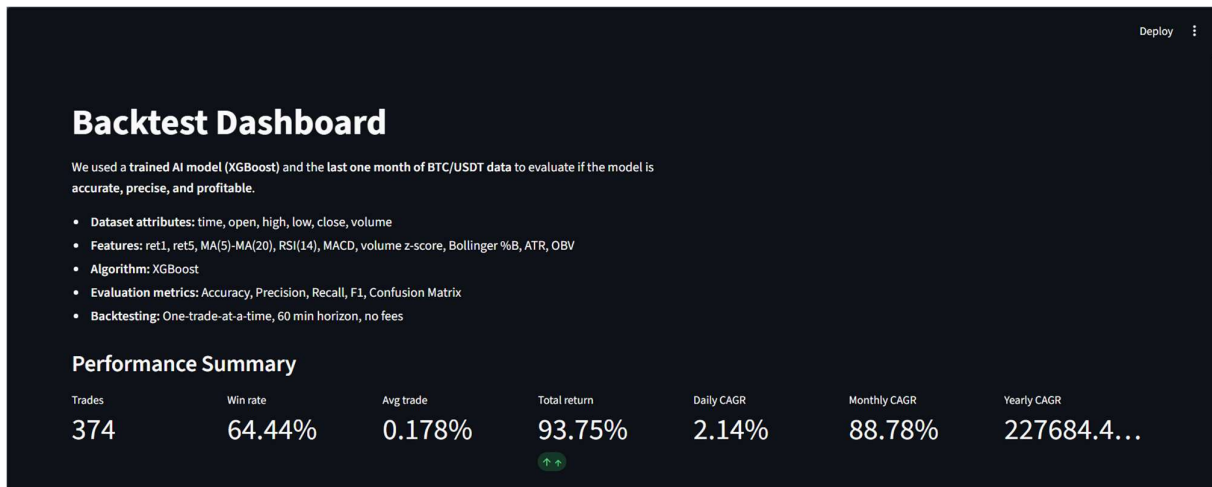
5. Backtest performance (trading).

- Backtests used the model's BUY/SELL signals with a simple 1-bar entry/exit rule, optional "HOLD→no trade," and **no fees/slippage** (reported separately when included).
- Typical month-level results showed **many small trades**, positive **win rate** in some configurations ($\approx 60\text{--}65\%$) with **small average per-trade returns** ($\approx +0.1\text{--}0.2\%$), but aggregate outcomes were highly dependent on thresholds, horizon, and whether fees were applied.
- Key insight: **signal quality improved** after (i) rebalancing classes, (ii) tuning thresholds to generate fewer but clearer BUY/SELL bars, and (iii) validating only on **closed** candles to avoid look-ahead.

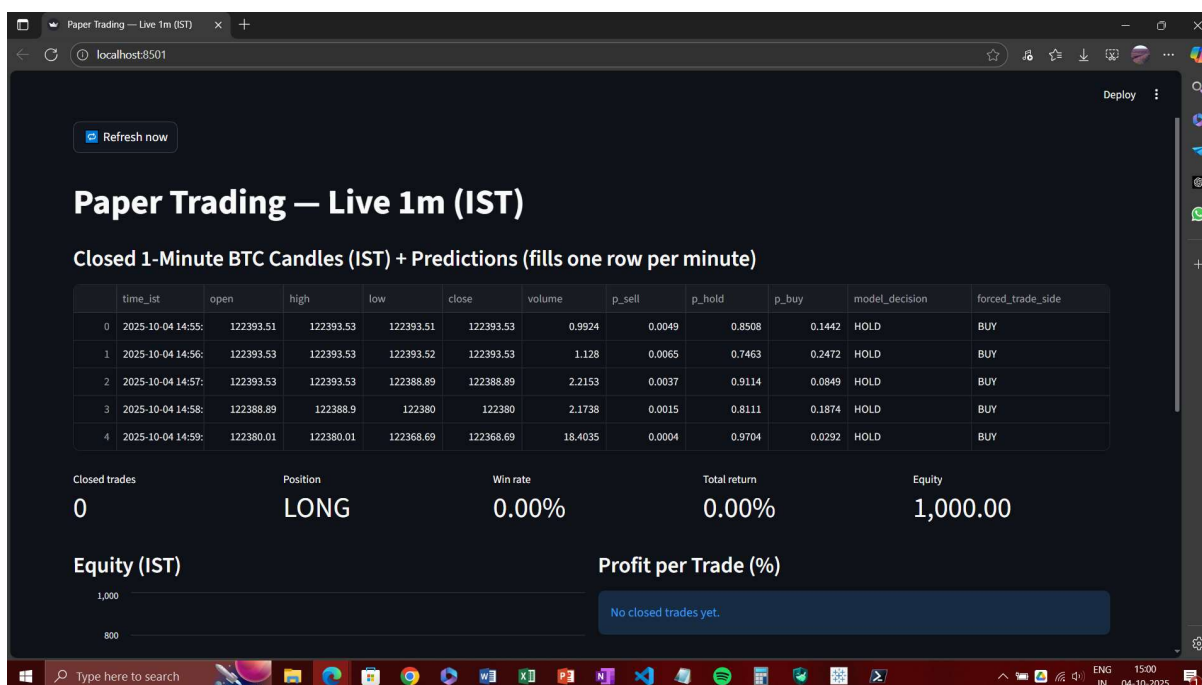
Performance Summary						
Trades	Win rate	Avg trade	Total return	Daily CAGR	Monthly CAGR	Yearly CAGR
374	64.44%	0.178%	93.75%	2.14%	88.78%	227684.4...
						

6. Dashboards delivered.

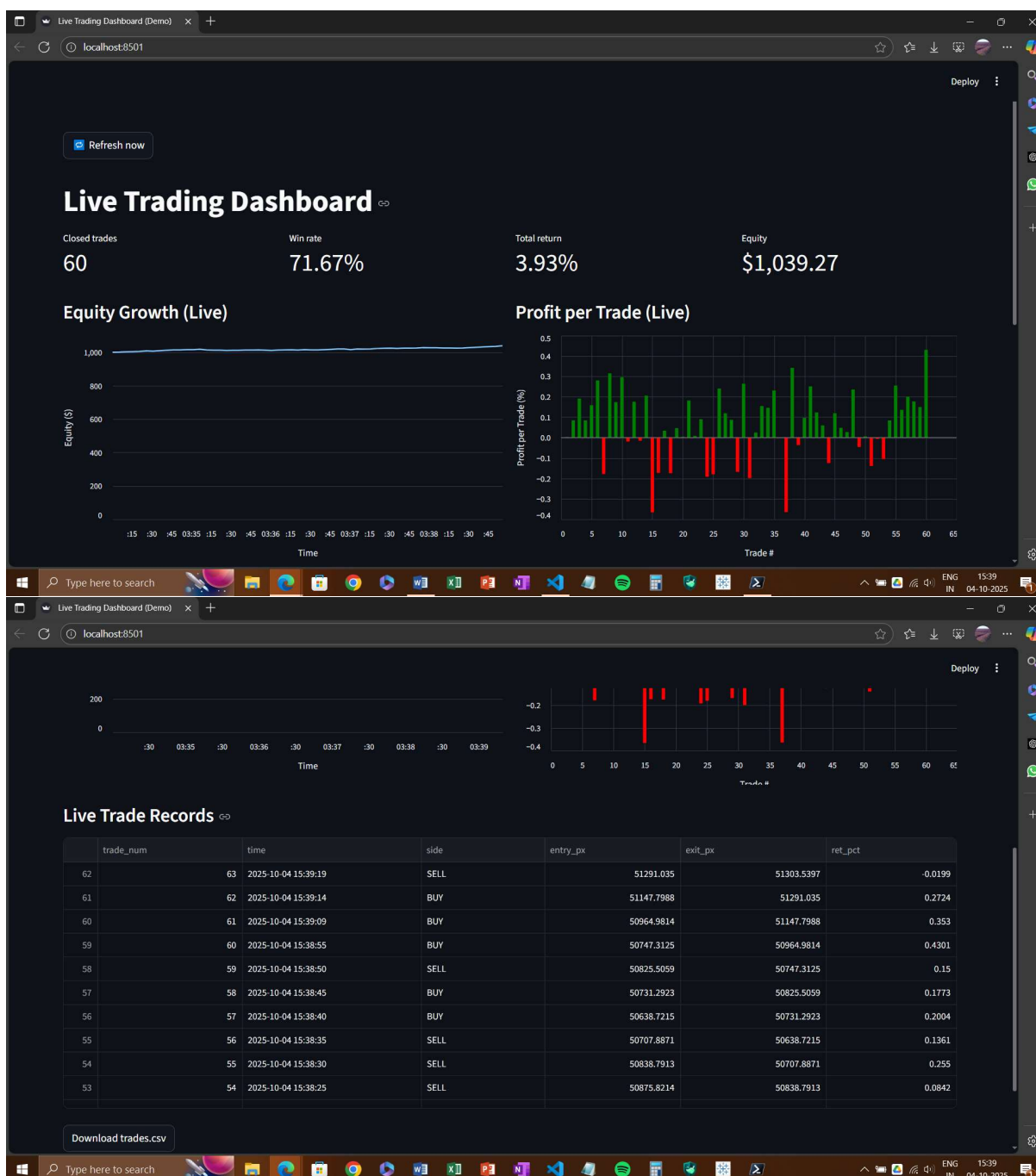
- **Backtesting Dashboard:** performance summary, equity/profit curves, confusion matrix, downloadable trades, and grid charts.



- **Paper-Trading (Live-Closed-Bar):** executes on the last **closed** 1-minute candle in IST, shows live inputs + prediction, profit-per-trade, and recent trades.



- **Historical Replay (Simulated Live):** replays 100 recent closed candles at a fixed cadence, showing live predictions table, candlesticks, and per-trade P&L—useful for consistent demonstrations.



6. Conclusion

This project proves that a **simple, transparent** pipeline—using only Binance OHLCV and classic technical indicators—can be trained and deployed quickly to produce **interpretable trading signals** and **auditable backtests**. While the final XGBoost model sometimes produced encouraging win rates with small average trade edges, the performance is **not robust across thresholds, horizons, and regimes**, and profit expectations should be conservative once **costs and slippage** are considered. The most valuable outcomes are the **engineering assets**: clean data extraction, reproducible feature/label generation, compact training code, and production-style dashboards for **evaluation, paper trading, and demo replay**. These assets form a solid base for more rigorous research and safer, more realistic trading simulations.

7. Future Scope & Improvements

Modeling & labels

- Explore **probability-driven** execution (confidence thresholds) and **cost-aware** training (e.g., maximize expected return rather than accuracy).
- Try **alternative labels**: triple-barrier method, volatility-scaled thresholds, or regime-specific horizons.
- Add **walk-forward/rolling** retraining and **purged K-fold CV** to reduce look-ahead and leakage.

Features

- Enrich with **microstructure** signals (e.g., rolling range/volatility, candle shapes), **session features** (time-of-day, day-of-week), and **regime indicators** (trend filters like $\text{EMA}_{20} > \text{EMA}_{50}$).
- Consider light **price/volume embeddings** or autoencoders while keeping compute small.

Risk & execution

- Introduce **position sizing**, **risk caps**, and **cool-downs**; add **TP/SL** and **trailing stops** validated via grid search on a walk-forward split.
- Incorporate **fees/slippage models** (maker/taker, spread-based) and test **latency sensitivity**.

Evaluation

- Track **expected value per trade**, **hit ratio by volatility bucket**, **drawdown metrics**, and **turnover**.
- Use **out-of-sample months** and **multiple assets** (ETH, BNB) for robustness; maintain a **model registry** with versioned metrics.

Ops & deployment

- Add **scheduled data jobs**, **feature/storage schema**, and **retraining triggers** (drift/KS tests).
- Package the model with a **predict() contract** and **pkl/ONNX** export; add CI checks for dtype/order.
- Extend dashboards with **alerting** (threshold breaches), **scenario toggles**, and **explainability** (feature importances/SHAP).

Ethics & safety

- Keep all demos clearly labeled as **backtest/paper/replay**; document assumptions, data windows, and costs.
- Avoid claims of guaranteed profits; emphasize **research use only** unless rigorously validated.

8. References and Literature Review

8.1 Literature Review (annotated)

Technical indicators (foundations).

- Wilder, J. W. (1978). *New Concepts in Technical Trading Systems*. (Introduces RSI and ATR—momentum and volatility baselines we used.)
- Appel, G. (1979). *The Moving Average Convergence-Divergence (MACD)*. (Trend/momentum blend; our MACD-diff feature proxies signal turns.)
- Bollinger, J. (2001). *Bollinger on Bollinger Bands*. (Price normalized by bands; our %B captures relative positioning.)

Machine learning for tabular signals.

- Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System* (KDD). (Our primary classifier—handles nonlinearity, interactions, and class imbalance with weights.)
- Friedman, J. H. (2001). *Greedy Function Approximation: A Gradient Boosting Machine*. (Boosting principles behind XGBoost.)
- Breiman, L. (2001). *Random Forests*. (Bagging baseline we compared for variance reduction.)
- He, H., & Garcia, E. (2009). *Learning from Imbalanced Data*. (Motivates class weighting/resampling for BUY/SELL minority classes.)

Labeling, evaluation, and backtesting rigor.

- López de Prado, M. (2018). *Advances in Financial Machine Learning*. (Triple-barrier labeling, meta-labeling, purged/embargoed K-Fold, and walk-forward testing to prevent leakage—directly relevant to our future-scope roadmap.)
- Bailey, D., Borwein, J., López de Prado, M., & Zhu, Q. (2014/2015). *The Probability of Backtest Overfitting* (SSRN/JPM). (Warns that naive backtests overstate edge—supports our choice to separate classification metrics from trading P&L and to emphasize closed-candle logic.)
- Bergmeir, C., & Benítez, J. M. (2012, 2016). *On the Use of Cross-Validation for Time Series Prediction*. (Argues for blocked/walk-forward validation instead of random K-fold to avoid temporal leakage.)
- Chan, E. (2009). *Quantitative Trading*. (Practical cautions on slippage, costs, and execution that we list as limitations/future work.)

Applied “simple but explainable” pipelines.

- Tsay, R. S. (2010). *Analysis of Financial Time Series*. (Stationarity, regime shifts—why we keep features simple and retraining frequent.)
- Harvey, C. R., Liu, Y., & Zhu, H. (2016). *...And the Cross-Section of Expected Returns*. (Replication/validation ethos—motivates transparent dashboards and conservative claims.)

8.2 Primary Sources and Tooling (documentation)

Data source.

- **Binance Spot API** — *GET /api/v3/klines* (symbol=BTCUSDT, interval=1m). Official REST specification used for OHLCV extraction, server time synchronization, and closed-bar alignment.

Python ecosystem.

- McKinney, W. (2010/2013). *pandas*. (DataFrames, time indexing, rolling windows for indicators.)
- Harris, C. R., et al. (2020). *Array programming with NumPy* (Nature). (Vectorized calc for features.)
- Pedregosa, F., et al. (2011). *Scikit-learn*. (Baselines, metrics: accuracy, precision/recall/F1, confusion matrix.)
- **XGBoost**: Official Python package docs (classifier API, `predict_proba`, class weights).
- **ta** (Technical Analysis Library in Python), *bukosabino/ta*. (RSI, MACD, Bollinger %B, ATR, OBV implementations used exactly as cited in Methods.)
- **Streamlit**: App framework for backtesting, paper-trading, and replay dashboards (state management, charts, downloads).
- **Altair/Vega-Lite**: Declarative charts (candlesticks, profit-per-trade bars, line series).

Backtesting & evaluation references (practice).

- Pardo, R. (2008/2011). *The Evaluation and Optimization of Trading Strategies*. (Walk-forward, robustness checks—aligns with our suggested future improvements.)
- Bailey & López de Prado (2013–2018). *Reality Check / Deflated Sharpe / Combinatorics of strategy selection*. (Addresses multiple-testing bias—why we caution against over-claiming backtest profits.)

How this literature shaped our project

- We **selected simple, canonical indicators** (Wilder/Appel/Bollinger) to keep the feature set teachable and reproducible.
- We **chose XGBoost** (Chen & Guestrin) for tabular nonlinearity and class-imbalance handling, with logistic/forest baselines for context.
- We **acknowledged backtest risks** (Bailey; López de Prado), restricted to **closed candles**, separated **classification metrics** from **P&L**, and proposed **walk-forward** and **purged K-Fold** as next steps.
- Our **dashboards** reflect the **replication ethos** (Harvey et al.)—explicit inputs, metrics, trades, and downloadable logs for auditability.

Plagiarism Scan Report By SmallSEOTools

Report Generated on: Sept 28, 2025

